Second-order mimetic velocity reconstruction with buildings in `palm_ftle`: first- versus second-order FTLE

pv_ftle — branch `palm2ndorder`

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Abstract

`palm_ftle` now reconstructs the PALM wind with second-order mimetic elements (`--order 2`, the new default) in a new double-precision C++ integrator. Each velocity component is quadratic across the cell and linear along each face, while every face keeps its exact flux. Buildings need special care, because a second-order reconstruction draws on neighbouring faces: faces are classified as fluid, wall or solid from PALM's fill values; a wall carries zero flux and no tangential slope; slopes and curvatures use fluid data only; and the velocity inside a building is zero. Against exact trajectories of an analytic divergence-free wind around three buildings, the second-order reconstruction is 2.5–9 times more accurate than the first-order one near the buildings and 5–50 times away from them, and no particle ends up inside a building. On a PALM large-eddy simulation of a built-up area (4 m grid, $T = -10$ s), the two orders give the same FTLE structures everywhere. The differences are largest in the lowest layer (4 m to 8 m), where a third of the cells are buildings and the first-order FTLE is biased low by 9% of the FTLE scale, and they fall to a median of 2% above 30 m. The differences are more than 100 times larger than the time-stepping error, so they come from the reconstruction.

1 Introduction

`palm_ftle` computes the finite-time Lyapunov exponent (FTLE) from PALM output. Particles are seeded at the corners of a window of the rectilinear PALM grid, advected over a time T (negative for backward FTLE) with RK4, and the FTLE is computed in each cell from the gradient of the flow map. Until now the wind between the faces was reconstructed with the lowest-order Raviart–Thomas (RT_0) element: each component is linear between its two faces and constant along each face, so the tangential wind jumps from one cell to the next. `wrf_ftle` already offers a second-order element. This report describes how it was carried over to PALM, where the main new difficulty is buildings, and compares the FTLE computed with the two orders.

2 Second-order reconstruction

PALM stores the velocity normal to each face of an Arakawa C grid: u on the x faces, v on the y faces and w on the z faces. Let $(\xi, \eta, \zeta) \in [0,1]^3$ be the local coordinates of a point in cell (i, j, k), and consider u (the other components are analogous). With u_0 and u_1 the values on the west and east faces, the first-order reconstruction is

$$u^{(1)} = (1-\xi)\, u_0 + \xi\, u_1. \tag{1}$$

The second-order ($Q_{2,1,1}$) reconstruction adds a linear variation along each face and a quadratic term across the cell:

$$u^{(2)} = (1-\xi)\, \tilde{u}_0 + \xi\, \tilde{u}_1 + \tfrac{1}{2}\, c\, \xi(\xi-1), \qquad \tilde{u}_f = u_f + s_f^{\eta}\,(\eta - \tfrac{1}{2}) + s_f^{\zeta}\,(\zeta - \tfrac{1}{2}). \tag{2}$$

- The tangential slopes s_f^{η}, s_f^{ζ} of face f are the central differences $\frac{1}{2}(u_{f+1} - u_{f-1})$ of the neighbouring faces in the y and z directions.

- The curvature c is the second difference $u_{-1} - 2u_0 + u_1$ (or $u_0 - 2u_1 + u_2$) along x, averaged over the two faces of the cell.

The slope terms have zero mean over each face, and $\xi(\xi - 1)$ vanishes on both faces, so every face keeps its exact flux and the reconstruction stays mimetic. The quadratic term lets $\partial u/\partial x$ vary across the cell, which the first-order element cannot do. On a uniform grid the reconstruction is exact for any linear wind and is continuous across faces. The slopes are index-space differences, so on PALM's stretched upper layers (4 m below 56 m, 8 m above) the reconstruction is no longer exact for a linear wind, but it remains much more accurate than the first-order one (unit test `test_order2_much_more_accurate_on_stretched_layers`).

3 Buildings

The second-order terms use the faces adjacent to the cell's own faces, so next to a building they would draw on values inside it. PALM writes the fill value -9999 on every face that touches a building; with `--zero-fill` (the default) these become 0. Using such faces as data would treat the inside of a building as still air, giving spurious tangential slopes along walls. The second-order reconstruction therefore follows the rules below.

Building mask. A cell is solid when all six of its faces are masked. The fill mask is constant in time; for the file used here, a face is masked exactly when one of its two cells is solid, so the face masks and the cell mask carry the same information. The ground face that `palm_ftle` adds below `zw_xy` is never masked, so a solid cell in the lowest layer is recognised from its other five faces.

Face types. Each face is

- *fluid* if neither of its cells is solid;
- a *wall* if exactly one of its cells is solid — the normal velocity on it is exactly 0;
- *solid* if both of its cells are solid.

Rules (order 2).

1. A wall face carries zero flux and no tangential slope: the wind normal to a wall is zero over the whole wall, including next to edges and corners.
2. The slopes of a fluid face use the neighbouring faces that are not solid. A wall's 0 counts as data. The difference is central when both neighbours are usable, one-sided when only one is, and 0 when neither is.
3. The curvature uses second differences at fluid faces only, again with a wall's 0 counted as data; there is no curvature from a wall face.
4. The velocity inside a solid cell is zero.

No no-slip condition is imposed: the tangential wind next to a wall comes from the fluid data, as in the LES, where the wind is resolved only down to the first grid point off the wall. A component parallel to a wall, such as the vertical wind in the first fluid cell beside a building, is reconstructed exactly as in open air, because its faces are fluid faces. Figure 1 shows the classification on a vertical section.

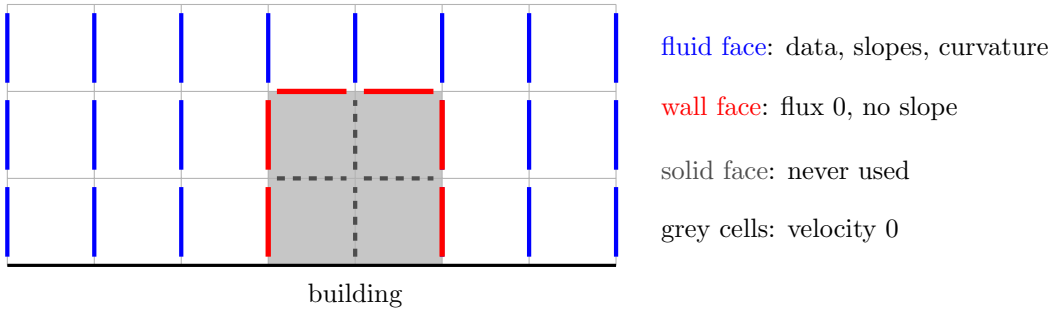


Figure 1: Face classification on an x–z section through a building two cells wide and two cells tall (only the x faces of fluid cells and the faces around the building are drawn). The slope along z of the fluid x face left of a wall uses its fluid neighbours; the curvature of u in the cell left of the wall uses the second difference at that cell's west face, in which the wall's 0 counts as data.

Particles. A seed lying exactly on a wall (all seeds on building surfaces, since seeds sit on grid corners) is assigned to the fluid cell on the other side of the wall, so it moves with the air beside the building instead of staying put. If an RK4 step ends inside a building, each coordinate whose crossing alone leads into the building is clamped just short of the wall, so the particle slides along it; if it is still inside, the step is undone. The first-order reconstruction uses the same mask, rules 1 and 4 and the same particle handling; rules 2 and 3 do not apply to it.

4 Other changes

- **New integrator.** `integrate_rk4_rectilinear` in `cpp/ftlecpp.cpp` (OpenMP, one fused RK4 loop per particle) replaces the old kernel for `palm_ftle`. Positions, times and axes are in double precision; the face data stay in single precision.
- **Time interpolation.** The old kernel assumed evenly spaced output times. The PALM file's times are 0.96–1.07 s apart, so after eight outputs the old kernel was 0.065 s out. The new integrator interpolates on the actual time axis. With frozen wind, the old kernel and the new first-order integrator agree to 0.2 mm (median). `palm_ftle_idx` still uses the old kernel.
- **Command line.** `--order 1|2` (default 2); `--visualise`, `--level`, `--cmax` open the same interactive level viewer as `wrf_ftle`, with buildings in grey.
- **Tests.** `tests/test_palm_reconstruction.py` (11 tests): exactness and continuity on a uniform grid, accuracy on stretched layers, zero velocity inside buildings, zero normal velocity everywhere on walls and roofs, seeds on walls, particles kept out of buildings, and recovery of the building mask from PALM-style face masks. CI runs the two PALM cases with order 2 and a third with order 1.

5 Analytic benchmark

The script `scripts/bench_palm.py` builds a divergence-free wind around three box buildings (24 m, 40 m and 16 m tall) in a 320 m × 320 m × 64 m domain:

- a sheared mean flow at 25° to the grid;
- horizontal vortices;
- an x–z roll ($|w| \lesssim 0.5$ m/s).

The stream functions are multiplied by a factor that vanishes on the buildings, so the walls and roofs are streamlines and the air slides along them, as for the resolved wind next to walls in an

LES ($R = 12\,\mathrm{m}$ wall layer). Face values are exact face means; exact trajectories are integrated with $\Delta t = 0.01\,\mathrm{s}$. The 5625 seeds, 8 m apart, are integrated over $T = -20\,\mathrm{s}$ (median displacement 54 m) at 8 m, 4 m and 2 m resolution. Seeds on or in buildings, where the analytic flow map is degenerate, are left out. "Near" means seeds within 8 m and FTLE cells within 16 m of a building.

	order 1			order 2		
grid spacing (m)	8	4	2	8	4	2
position error near, median (m)	5.2	1.4	0.43	2.1	0.28	0.047
position error near, 95th pct (m)	36	19	9.7	18	4.3	1.4
position error away, median (m)	1.0	0.33	0.12	0.13	0.016	0.0024
FTLE error near, median (%)	7.4	1.8	0.57	2.9	0.42	0.081
FTLE error away, median (%)	1.5	0.47	0.16	0.29	0.032	0.0047

Table 1: Analytic benchmark: errors against exact trajectories. FTLE errors are relative to the FTLE scale (99th percentile of the exact FTLE, $0.11\,\mathrm{s}^{-1}$). No particle ends inside a building with either order.

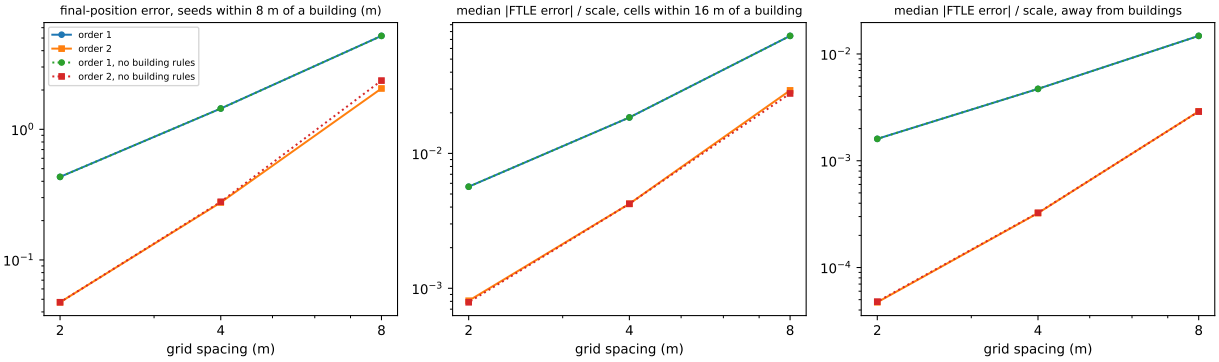


Figure 2: Analytic benchmark: median errors against grid spacing. Order 1 converges at first order, order 2 at second order. The dotted lines are order 2 without the building rules (all faces used as data, walls included); order 1 is unaffected by them.

Table 1 and Figure 2 show:

- Order 2 converges at second order, order 1 at first order.
- Order 2 is 2.5–9 times more accurate near buildings and 5–50 times away from them.
- The building rules improve the tails at the coarsest grid (95th percentile 18 m instead of 19 m, median 2.1 m instead of 2.4 m). Without them, one particle ends inside a building.
- With a no-slip variant of the wind, which vanishes on the walls, the rules make almost no difference: the zeros on the walls are then accurate data.

6 Comparison on PALM data

6.1 Set-up

The PALM file `small_blf_day_loc1_4m_xy_N04.003.nc` is a large-eddy simulation of a built-up area:

- horizontal grid spacing 4 m;
- 18 layers, 4 m thick up to 56 m and 8 m above, up to 96 m;

- the ground face is reconstructed at 4 m;
- output roughly every second.

The FTLE was computed:

- in the window $i = 20$–100, $j = 5$–95 (320 m × 360 m, 1.4×10^5 seeds, 129 600 cells);
- backwards over $T = -10$ s from time index 20 ($t = 7511$ s);
- with time-dependent wind;
- with 575 RK4 steps (CFL 0.25).

In the window, buildings fill 35% of the lowest layer (4 m to 8 m), 11% of the second and 0.6% of the third; none reaches 16 m. Differences are quoted relative to the FTLE scale $S = 0.24\,\mathrm{s}^{-1}$, the 99th percentile of the second-order FTLE. The scripts that produce the runs, statistics and figures are listed in the appendix.

6.2 Three layers

Figure 3 compares the FTLE at three layers:

- the lowest layer (4 m to 8 m), among the buildings;
- layer 8 (36 m to 40 m), in the middle;
- the top layer (88 m to 96 m).

Lowest layer. The two orders show the same structures: high FTLE in the street canyons and in the wakes of building blocks, and low FTLE in the open area. The differences, however, are large:

- median $|\Delta| = 8.5\%$ of S, 95th percentile 33%;
- correlation 0.80 (Figure 4);
- order 1 is lower in most cells, by 9% of S on average (15% of the layer's mean FTLE).

The largest differences are not confined to the cells next to buildings: they also cover patches of open ground one or more cells away from walls. This is the layer with the strongest vertical shear. Order 1 holds the horizontal wind constant with height within each layer, so the shear between the seeds at the bottom and top of a cell comes only from the jump between layers. Order 2 interpolates linearly with height, using a one-sided slope at the ground. The low bias matches the first-order deficit found in the lowest layer of WRF (see the `wrf_ftle` order-2 report).

Middle and top layers. There are no buildings at these heights and the agreement is close:

- correlation 0.98 and 0.97;
- median $|\Delta|$ 2.0% and 1.8% of S, 95th percentile about 10%;
- bias -0.9% and -1.4%.

The differences sit along the sharp FTLE ridges, where they are signed (often dipoles), so the ridges shift or sharpen slightly between the two orders but are not created or removed. At the top the layers are 8 m thick, so order 1 is coarser in the vertical and the ridges around $y = 280$ m to 340 m show the largest differences of that layer.

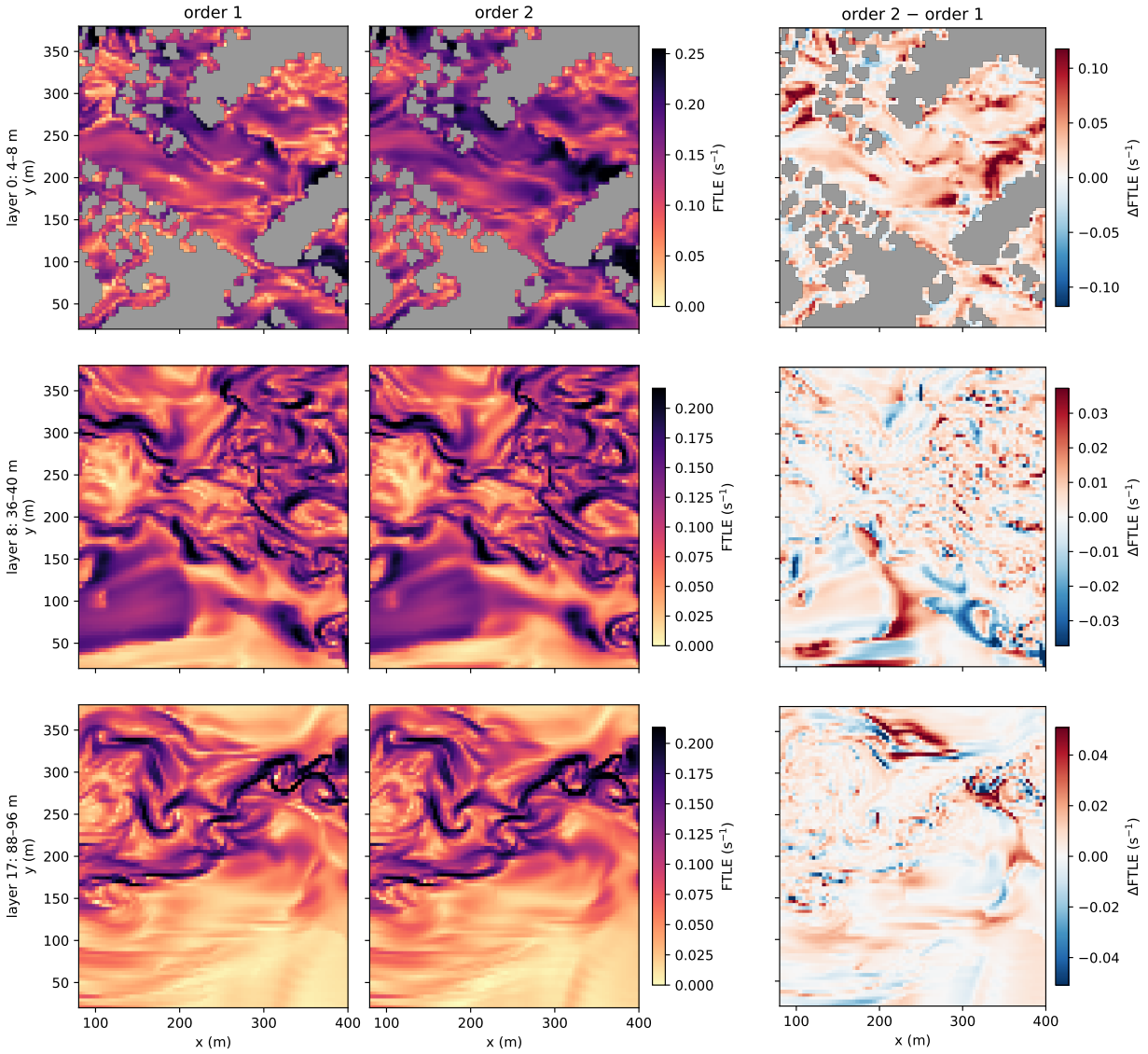


Figure 3: FTLE ($T = -10\,\mathrm{s}$) with order 1 (left) and order 2 (centre), and their difference (right), at the lowest, a middle and the top layer. Buildings are grey. Each row has its own colour scales; note that the difference scale of the lowest layer is three times that of the others.

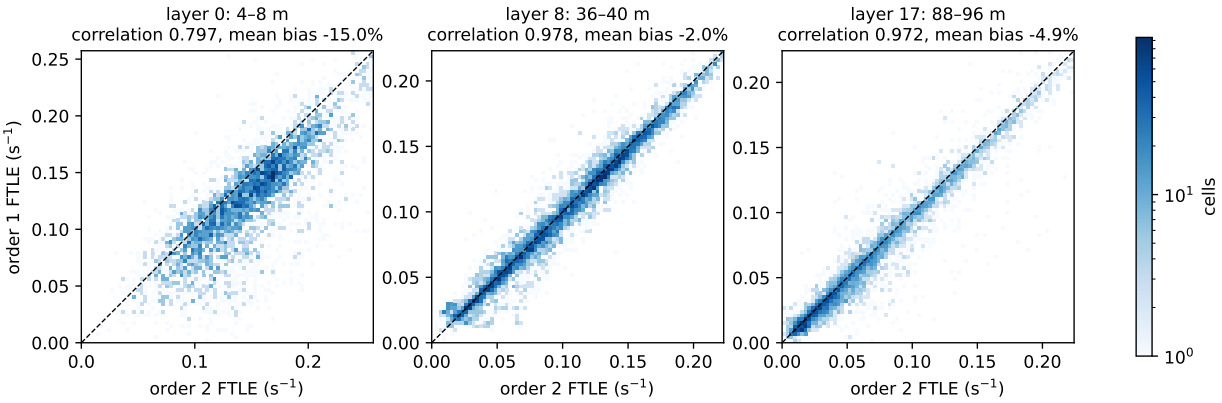


Figure 4: Order-1 against order-2 FTLE in every fluid cell of the three layers (log colour scale). The mean bias is relative to the mean order-2 FTLE of the layer.

6.3 Statistics by layer and distance to buildings

Figure 5 shows the median and 95th percentile of the difference in every layer:

- The median decreases steadily from 8.5% at the ground to 2% at 40 m, and stays there.
- The bias (order 1 lower) is concentrated in the lowest layer, with −2% in the second and third, and −1.4% to +0.1% above.

Table 2 groups the cells of the whole window by their distance to the nearest building:

- Next to buildings the order difference is three times larger than far from them.
- The building rules change the order-2 FTLE next to buildings by as much as switching from order 1 to order 2 (median 4.5% vs 5.6%). They make no difference farther than three cells away.

Figure 6 maps the effect of the rules in the lowest layer: it is confined to the cells adjacent to walls, and in most of them the FTLE is higher without the rules.

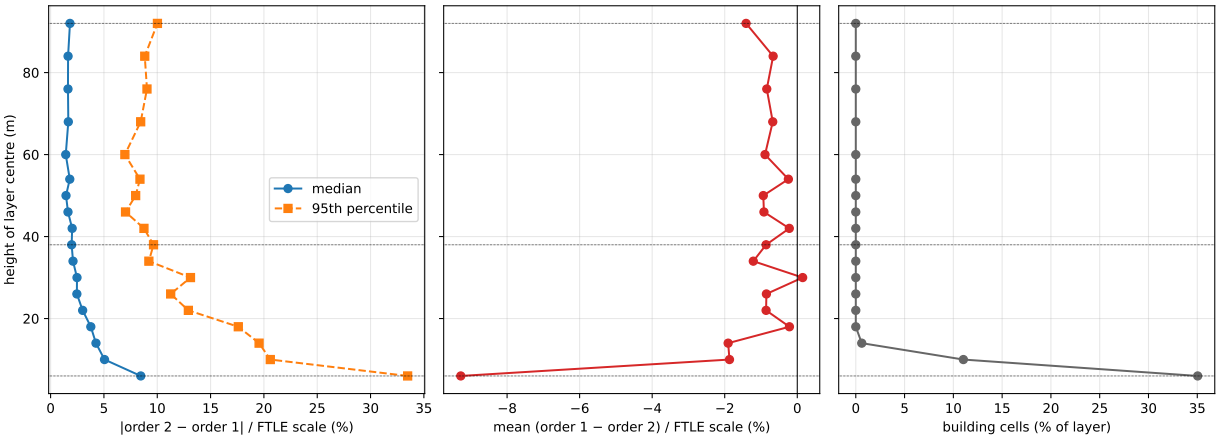


Figure 5: By layer: median and 95th percentile of |order 2 − order 1| (left), mean order 1 − order 2 (centre), both relative to S, and the fraction of building cells (right). Dotted lines mark the three layers of Figure 3.

6.4 Time step and seeds on faces

The last two columns of Table 2 check the numerics:

distance to building	cells	order 2 vs 1	rules on vs off	CFL/4 (o2)	nudge (o1 / o2)
1 cell (adjacent)	6226	5.6 / 23	4.5 / 21	0.03 / 1.9	2.4 / 1.8
2–3 cells	15147	4.1 / 19	0 / 3.0	0.004 / 0.03	0.013 / 0.003
farther	104866	2.0 / 11	0 / 0	0.002 / 0.014	0.006 / 0.002

Table 2: Median / 95th percentile of FTLE differences in % of S, by chessboard distance to the nearest building cell (in 3D), over all layers. "CFL/4": four times smaller time step. "Nudge": seeds displaced by 10 µm at random (median only).

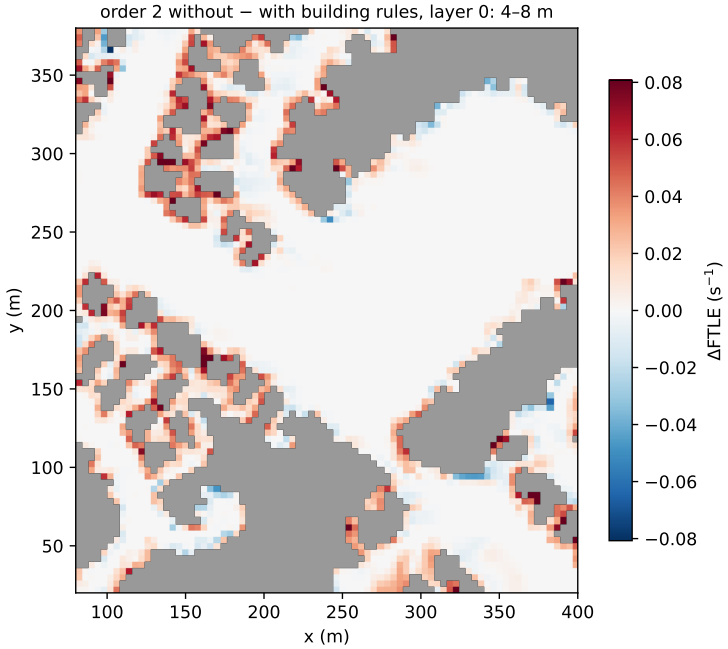


Figure 6: Effect of the building rules on the order-2 FTLE in the lowest layer: order 2 with every face used as data (walls and faces inside buildings) minus order 2 with the rules.

- **Time step.** Reducing the time step by four changes the FTLE by a median 0.002–0.03% of S, 150–1000 times less than the difference between the orders, so the comparison measures the reconstruction.
- **Seeds on faces, away from buildings.** The seeds lie exactly on cell faces, where the first-order tangential wind jumps; nudging them by 10 µm changes the order-1 FTLE three to four times more than the order-2 FTLE, whose wind is continuous across fluid faces.
- **Seeds on building walls.** Here both orders are sensitive (about 2% median): a seed nudged into the building stays there with zero velocity, while one nudged out moves with the air. The FTLE of cells whose corners lie on walls is ill-defined for either order and should be read with care.

7 Conclusions

- The second-order reconstruction in `palm_ftle` keeps the exact face fluxes, gives a continuous wind across fluid faces, and handles buildings by using fluid data only, with zero normal velocity on walls and zero velocity inside buildings.
- In the analytic benchmark it converges at second order, is 2.5–9 times more accurate than order 1 near buildings and 5–50 times away from them, and keeps all particles out of buildings.

- On PALM data, order 1 and order 2 agree closely above the buildings (median difference 2% of the FTLE scale, correlation 0.97–0.98). In the lowest layer they differ by 8.5% (median), and order 1 is biased low by 9%, consistent with its missing within-layer shear.
- Next to buildings the building rules matter as much as the order. Without them the second-order wind would be built partly from the zeros inside buildings.
- Order 2 is the default. Order 1 remains available (`--order 1`) for comparison.

A Reproducing the results

```
python scripts/bench_palm.py OUT FIG --slip      # analytic benchmark
python scripts/palm_order_compare.py small_blf_day_loc1_4m_xy_N04.003.nc CMP
python scripts/palm_order_report_figs.py small_blf_day_loc1_4m_xy_N04.003.nc \
       CMP report/palm_order2/figures
palm_ftle small_blf_day_loc1_4m_xy_N04.003.nc --imin=20 --imax=100 \
       --jmin=5 --jmax=95 --time-index=20 --tintegr=-10 --order=2 --visualise
```